Set 3B: Op-Amp Circuits with Diodes

ECEN 325 — Lecture 3 (Diodes) practice set

What this set trains. The same state method as set 3A, now with ideal op-amps. In each assumed state an op-amp either has negative feedback (virtual short) or has lost it because a diode in its feedback path is OFF; then it runs open-loop and its output swings until a diode turns on (Sample Problem 5). Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-3 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1/2 tools):

T1 *Diode models.* Ideal: OFF $\Leftrightarrow V_D < 0$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0$). Constant-voltage-drop (CVD): OFF $\Leftrightarrow V_D < 0.7\,\text{V}$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0.7\,\text{V}$ source). Exponential: $I_D = I_S e^{V_D/nV_T}$, i.e. $V_D = nV_T \ln(I_D/I_S)$, $V_T \approx 25\,\text{mV}$.

T2 *State method.* Assume ON/OFF for every diode, solve the now-linear circuit (KCL, Lecture 2), then check the assumptions; each check is a condition on V_i. The valid states tile the V_i axis.

T3 *Small signal.* Around a DC point I_D the diode is a resistor $r_d = \frac{nV_T}{I_D}$ (valid while $|v_d| \ll nV_T$). Capacitors: open at DC; $\frac{1}{sC}$ in AC; DC supplies are AC ground.

T4 *Rectifier with a filter capacitor* ($f = 60\,\text{Hz}$, $T = \frac{1}{f}$ half-wave, $\frac{1}{2f}$ full-wave): peak $V_p = \hat{V}_S -$ (diode drops in the path); ripple $V_r \approx V_p \frac{T}{RC}$; $I_{D,av} \approx I_L\left(1 + \pi\sqrt{2V_p/V_r}\right)$, $I_{D,max} \approx I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right)$; PIV = the largest reverse voltage a diode sees.

Steps:

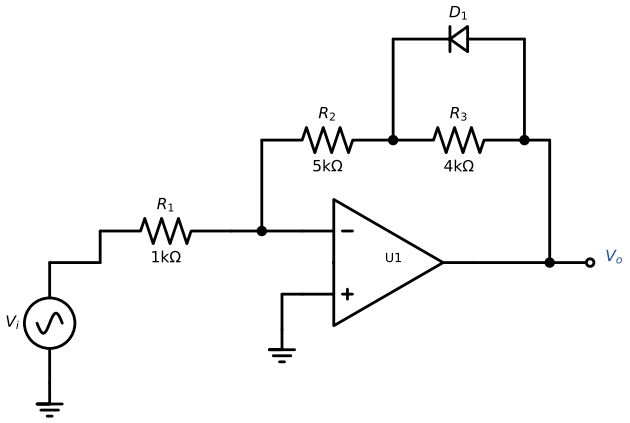
1. *Pick the model* stated in the problem.
2. *Transfer characteristic:* for each ON/OFF state: replace the diodes, solve, check $\Rightarrow$ range of V_i; then join the pieces (they must meet at the breakpoints) and sketch.
3. *Small signal:* DC operating point first, then r_d, then the AC circuit exactly as in Lectures 1–2, then add DC and AC.
4. *Rectifiers:* peak $\rightarrow$ ripple (or C for a given ripple) $\rightarrow$ diode currents $\rightarrow$ PIV $\rightarrow$ transformer ratio.

Problems

Easy problems

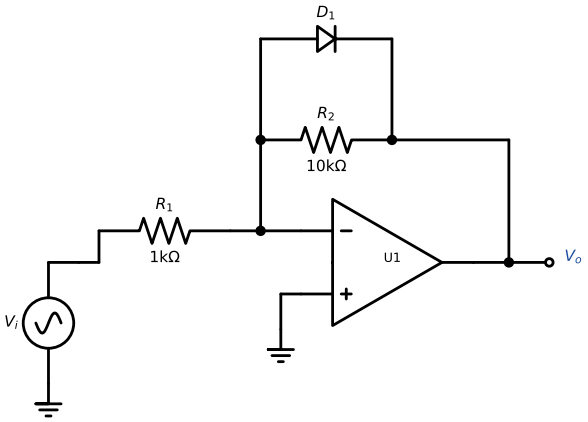
Problem 3B.1 **Easy**

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



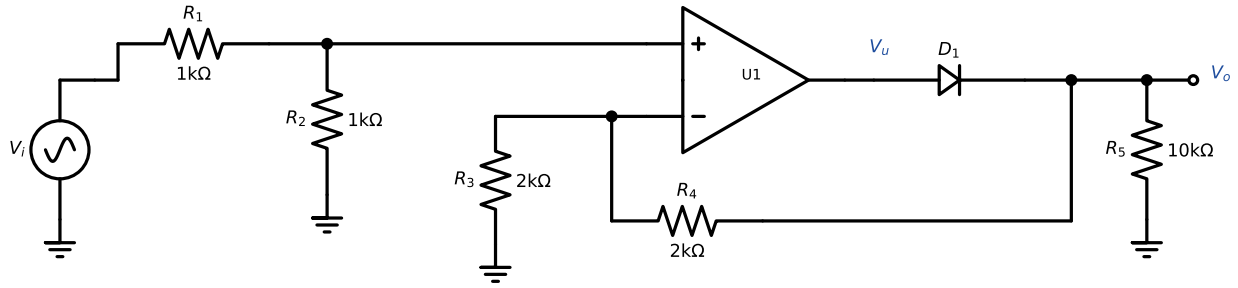
Problem 3B.2 **Easy**

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



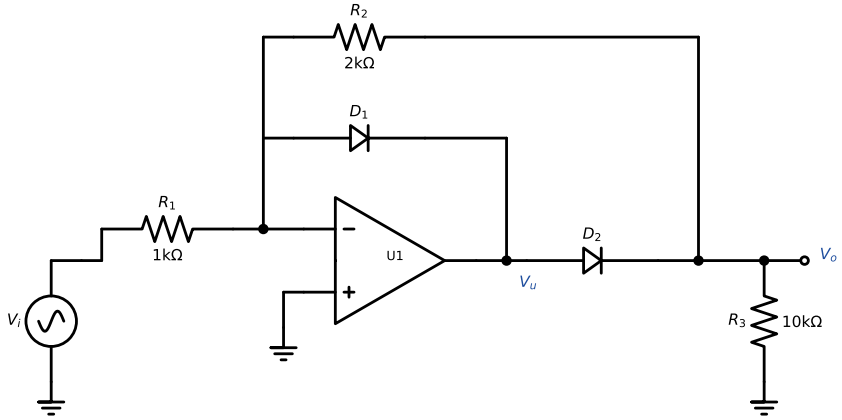
Problem 3B.3 — Easy

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 500\,t)$.



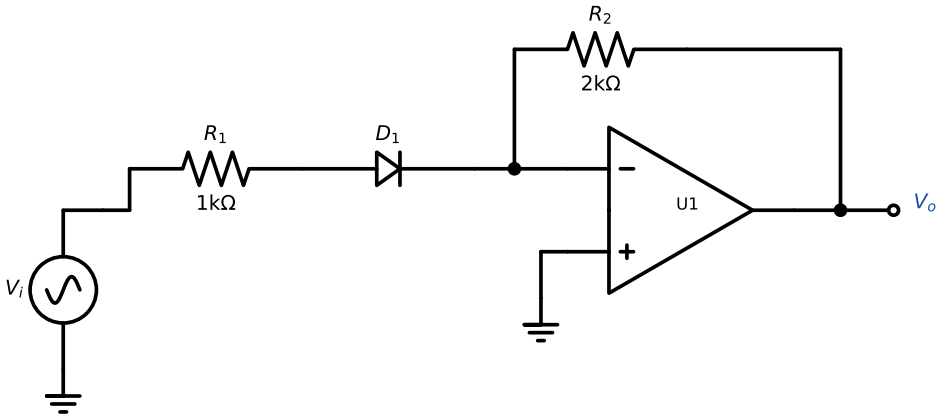
Problem 3B.4 — Easy

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



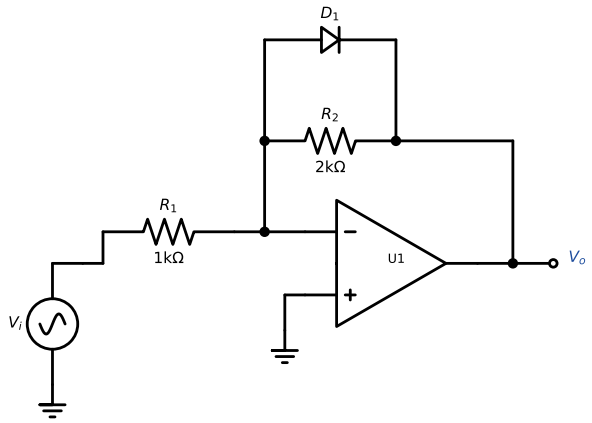
Problem 3B.5 — Easy

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$.



Problem 3B.6 **Easy**

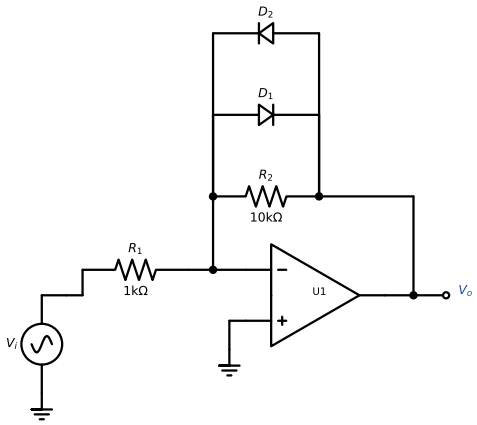
For the circuit below (ideal op-amps), use the ideal-diode model. Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



Medium problems

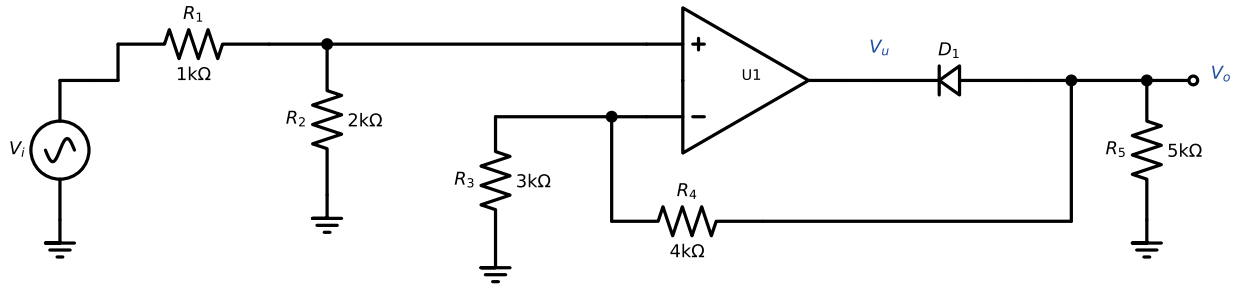
Problem 3B.7 **Medium**

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$.



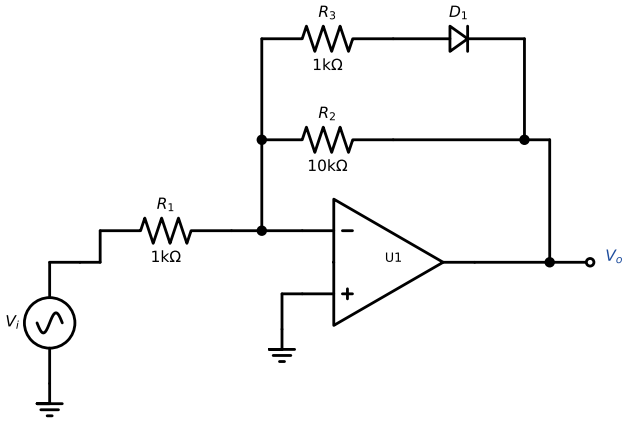
Problem 3B.8 — Medium

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 500\,t)$.



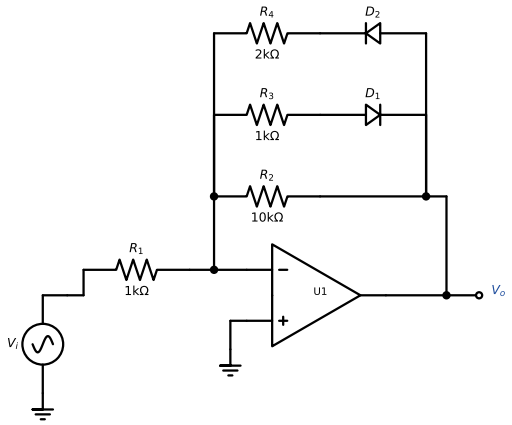
Problem 3B.9 — Medium

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



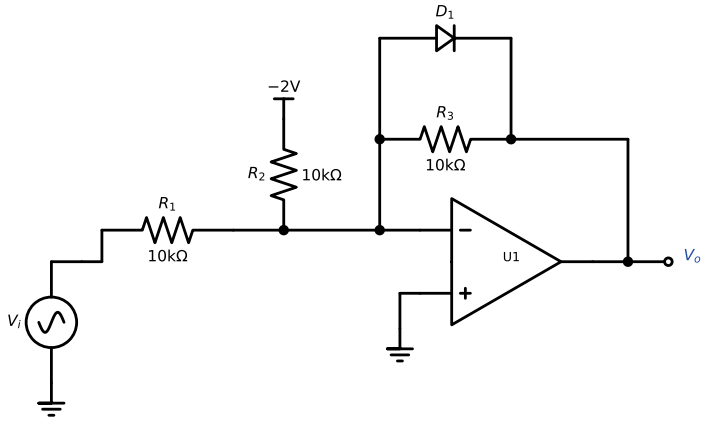
Problem 3B.10 — Medium

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



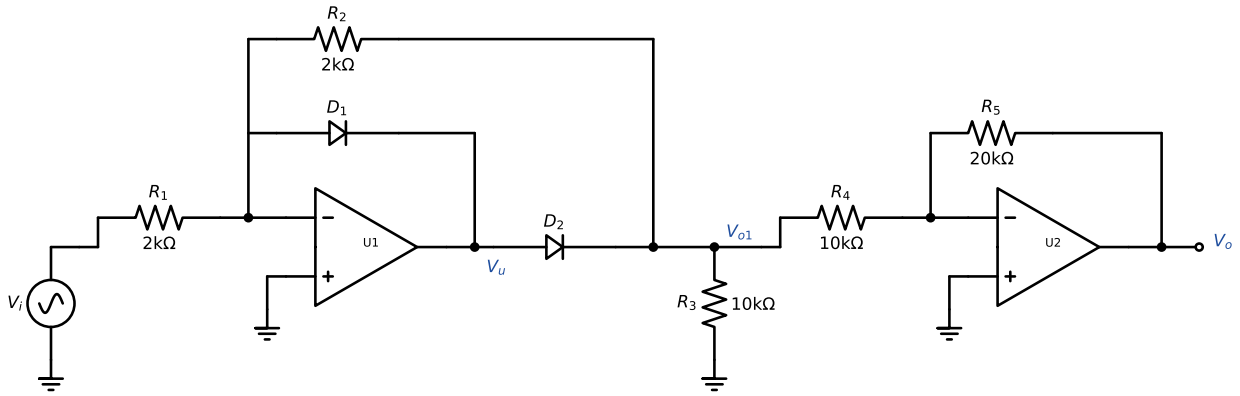
Problem 3B.11 — Medium

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 4\sin(2\pi \cdot 1000\,t)$.



Problem 3B.12 — Medium

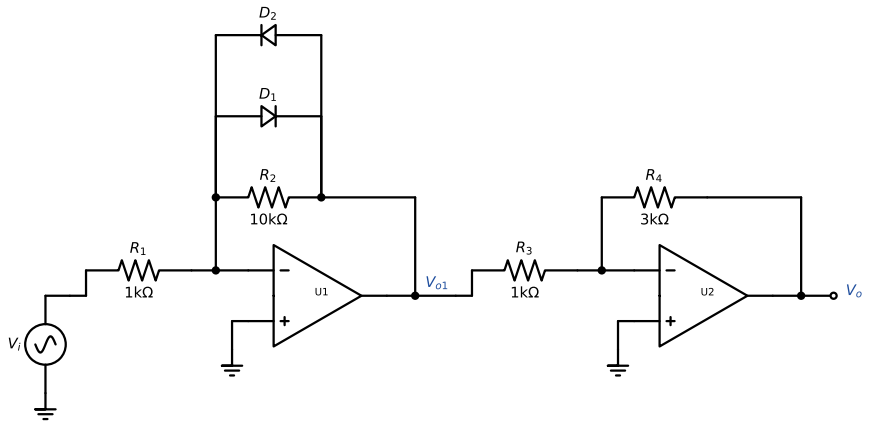
For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



Hard problems

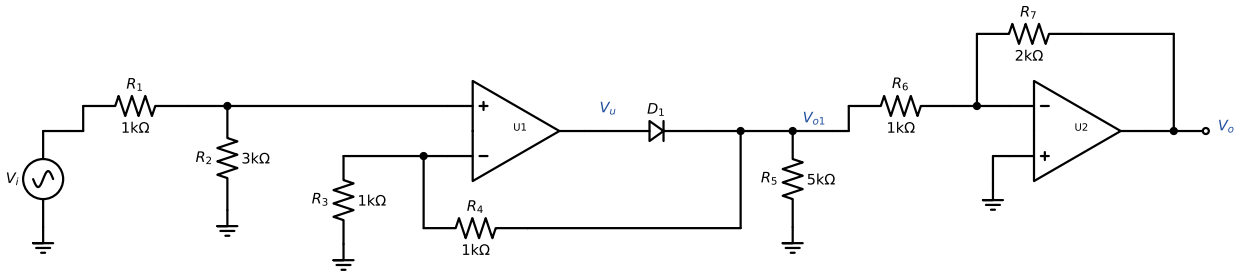
Problem 3B.13 — Hard

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 0.3\sin(2\pi \cdot 1000\,t)$.



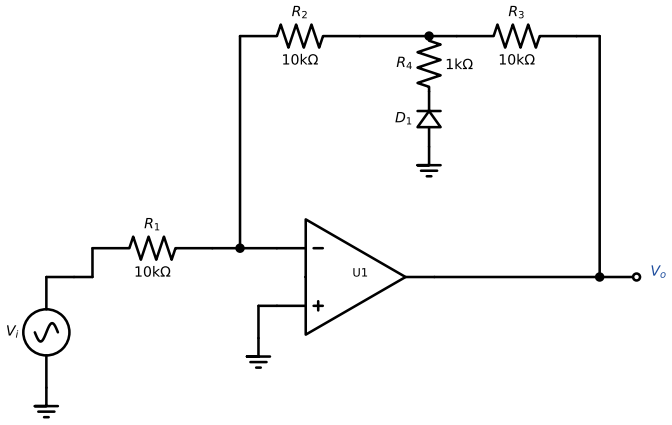
Problem 3B.14 — Hard

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$.



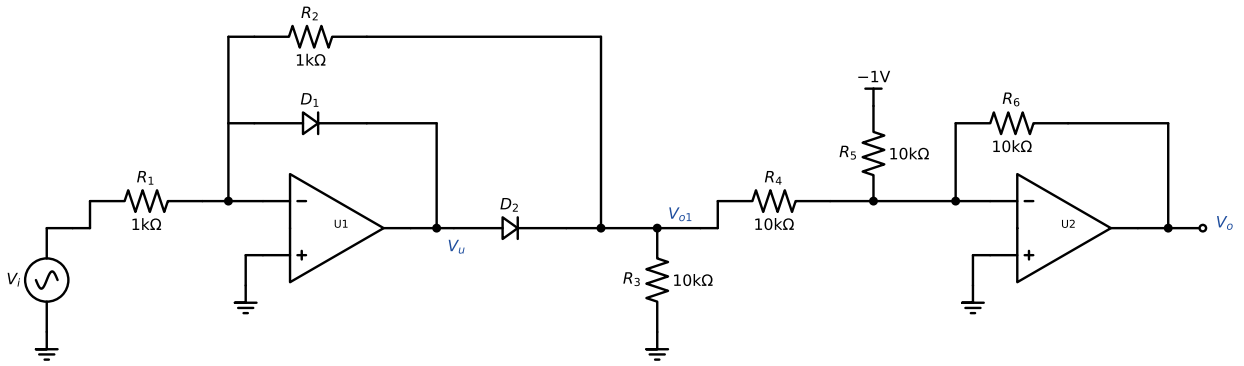
Problem 3B.15 — Hard

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



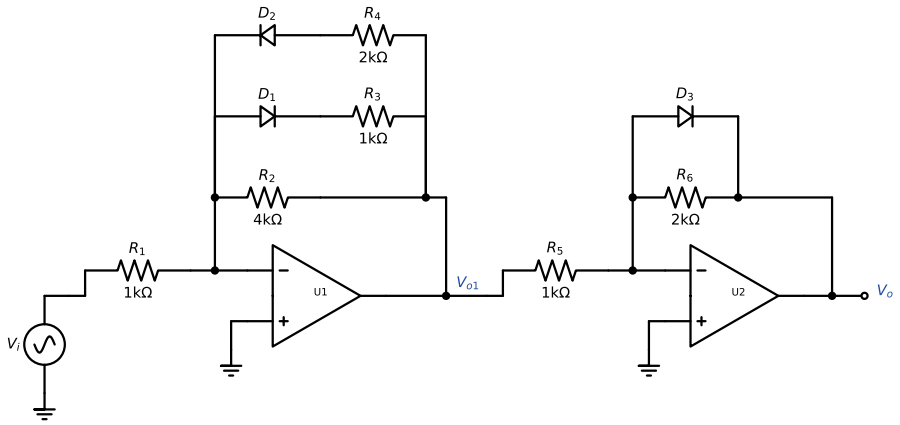
Problem 3B.16 — Hard

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



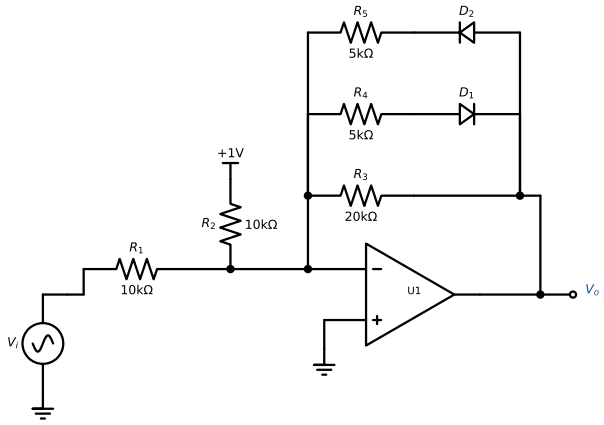
Problem 3B.17 — Hard

For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



Problem 3B.18 **Hard**

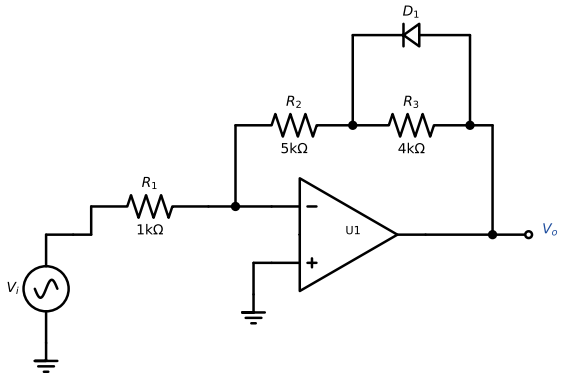
For the circuit below (ideal op-amps), use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$.



Solutions

Every solution follows the steps of the Lecture-3 pipeline card.

Solution 3B.1 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input; V_a = node joining D_1, R_2, R_3.

Step 2 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_o - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1} + \frac{0-V_a}{R_2} = 0 \\ &\text{KCL at } V_a: && \frac{V_a}{R_2} + \frac{V_a - V_o}{R_3} - I_{D_1} = 0 \\ &\text{D1 ON}: && V_o - V_a = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -5V_i + 0.7 \\ I_{D_1} &= -V_i - 0.175\ \text{mA} \end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < -0.175\,\text{V}$. So this state holds for $\boxed{V_i < -0.175}$, where $V_o = -5V_i + 0.7$.

Step 3 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0-V_i}{R_1} + \frac{0-V_a}{R_2} = 0$$

$$\text{KCL at } V_a: \qquad \frac{V_a}{R_2} + \frac{V_a - V_o}{R_3} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -9V_i$$
$$V_{D_1} = V_o - V_a = -4V_i$$

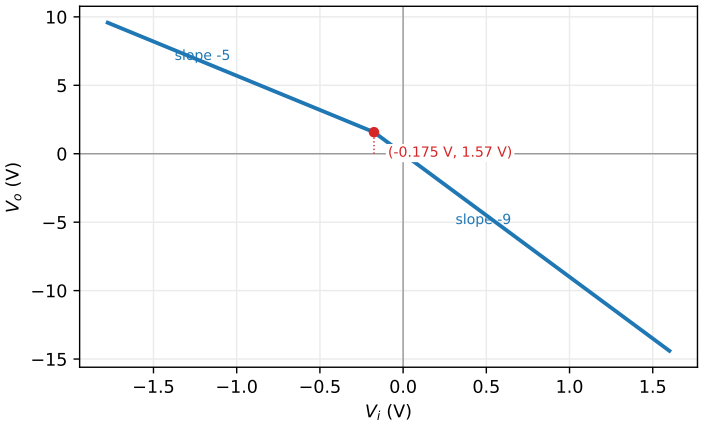
Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i > -0.175\,\text{V}$. So this state holds for $\boxed{V_i > -0.175}$, where $V_o = -9V_i$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} -5V_i + 0.7 & V_i < -0.175 \\ -9V_i & V_i > -0.175 \end{cases}$$

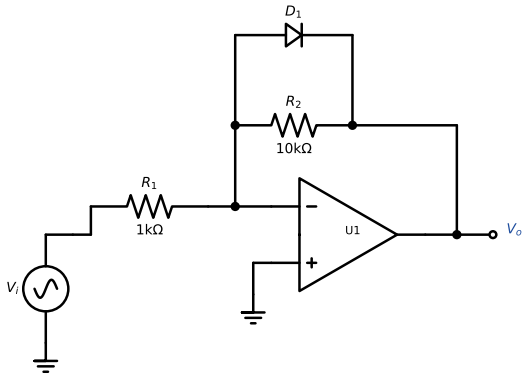
Check: the pieces must meet at each breakpoint — at $V_i = -0.175\,\text{V}$ both pieces give $V_o = 1.575\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} -5V_i + 0.7 & V_i < -0.175 \\ -9V_i & V_i > -0.175 \end{cases}$$

Solution 3B.2 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -10V_i$$
$$V_{D_1} = V_- - V_o = 10V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.07\,\text{V}$. So this state holds for $\boxed{V_i < 0.07}$, where $V_o = -10V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_- - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad 0 - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -0.7$$

$$I_{D_1} = V_i - 0.07 \text{ mA}$$

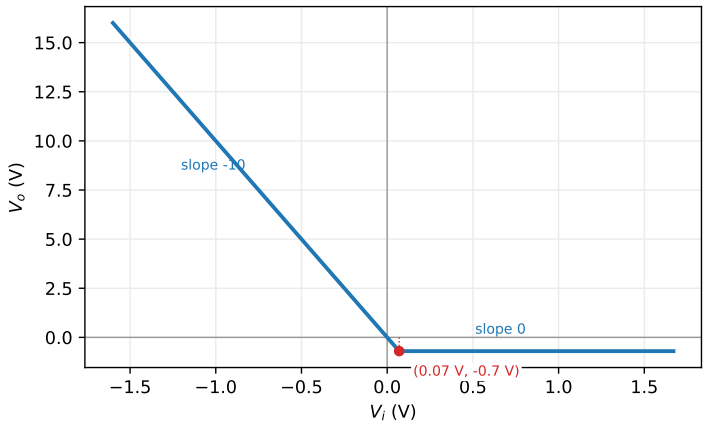
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.07\,\text{V}$. So this state holds for $\boxed{V_i > 0.07}$, where $V_o = -0.7$.

Step 4 — Assemble and sketch

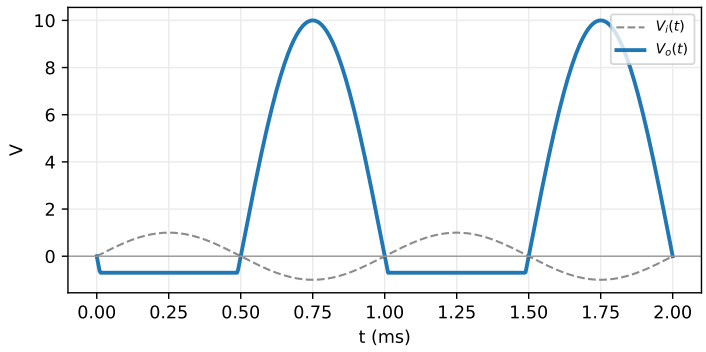
Put the valid states together:

$$V_o = \begin{cases} -10V_i & V_i < 0.07 \\ -0.7 & V_i > 0.07 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.07\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



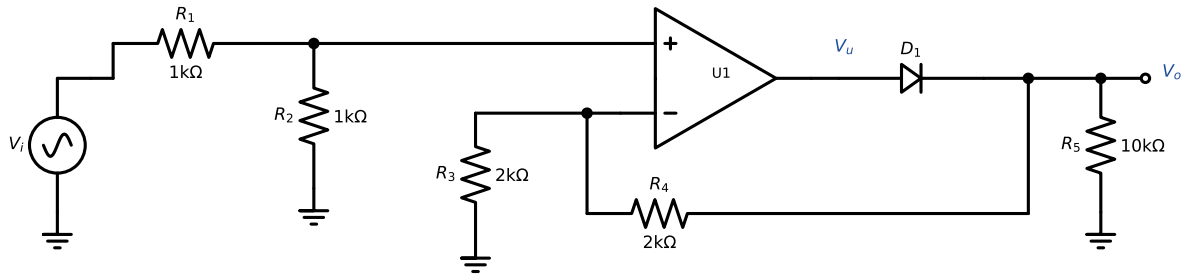
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -10V_i & V_i < 0.07 \\ -0.7 & V_i > 0.07 \end{cases}$$

Solution 3B.3 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_+ = the op-amp's + input; V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. With this assumption U1 has no feedback path from its output back to its − input: it runs *open-loop*, $V_u = A(V_+ - V_-)$ with $A \to \infty$, so its output is pushed to $\pm\infty$ (in practice, a supply rail) unless $V_+ = V_-$ exactly.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_+ : \quad \frac{V_+ - V_i}{R_1} + \frac{V_+}{R_2} = 0$$

$$\text{KCL at } V_- : \quad \frac{V_-}{R_3} + \frac{V_- - V_o}{R_4} = 0$$

$$\text{KCL at } V_o : \quad \frac{V_o - V_-}{R_4} + \frac{V_o}{R_5} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$V_{D_1} = V_u - V_o \approx A\,(0.5V_i)$$

Check the assumptions: $V_{D_1} < 0.7$; since $A \to \infty$ this needs $0.5V_i < 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = 0$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_u - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_+ : \quad \frac{V_+ - V_i}{R_1} + \frac{V_+}{R_2} = 0$$

$$\text{KCL at } V_- \,(= V_+) : \quad \frac{V_+}{R_3} + \frac{V_+ - V_o}{R_4} = 0$$

KCL at V_o :	$\dfrac{V_o - V_+}{R_4} + \dfrac{V_o}{R_5} - I_{D_1} = 0$
D1 ON :	$V_u - V_o = 0.7$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i$$
$$I_{D_1} = 0.35 V_i \text{ mA}$$

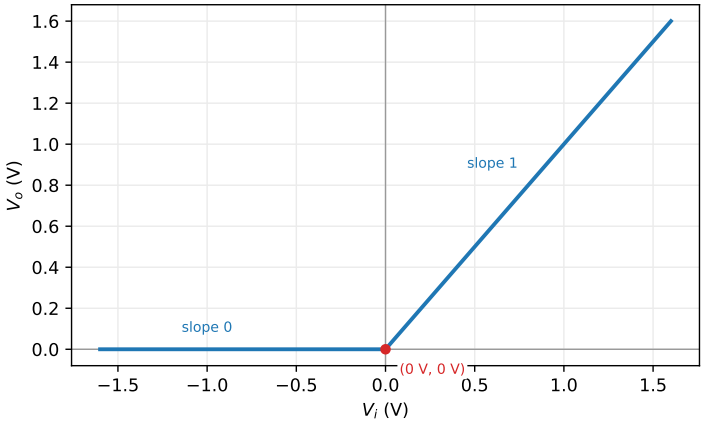
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$. So this state holds for $\boxed{V_i > 0}$, where $V_o = V_i$.

Step 4 — Assemble and sketch

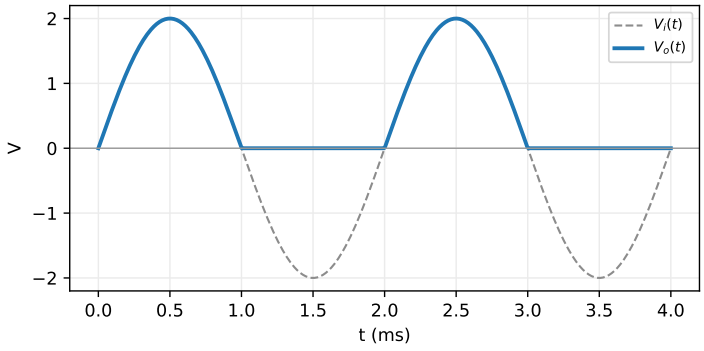
Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0 \\ V_i & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



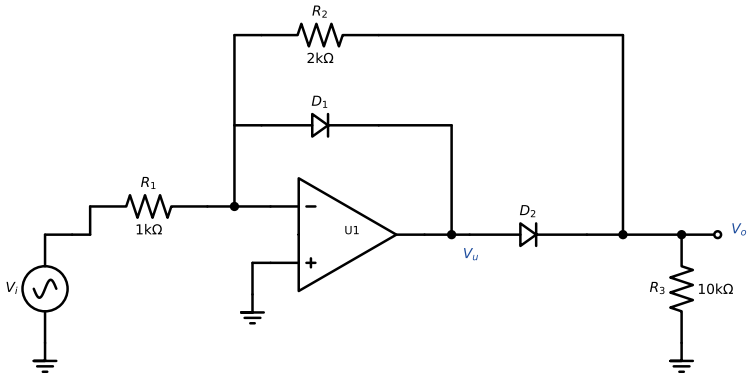
For $V_i(t) = 2\sin(2\pi \cdot 500\, t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0 & V_i < 0 \\ V_i & V_i > 0 \end{cases}$$

Solution 3B.4 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_u - V_o = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- \,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o}{R_2} + \frac{V_o}{R_3} - I_{D_2} = 0 \\ &\text{D2 ON}: && V_u - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -2V_i \\ V_{D_1} = V_- - V_u &= 2V_i - 0.7 \\ I_{D_2} &= -1.2V_i \text{ mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = -2V_i$.

Step 3 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_- - V_u = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1}+\frac{0-V_o}{R_2}+I_{D_1}=0\\
&\text{KCL at } V_o: && \frac{V_o}{R_2}+\frac{V_o}{R_3}=0\\
&\text{D1 ON}: && 0-V_u=0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0\\
I_{D_1} &= V_i \text{ mA}\\
V_{D_2} = V_u - V_o &= -0.7
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0$.

Step 4 — The other combinations

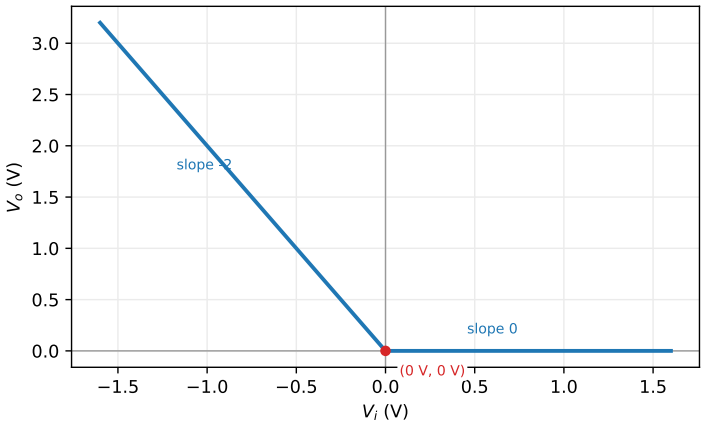
- D_1 OFF, D_2 OFF: $V_{D_1} < 0.7$ needs $V_i < 0$; $V_{D_2} < 0.7$ needs $V_i > 0$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 0.7$; $I_{D_2} > 0$ is never true — these cannot all hold, so this state never occurs.

Step 5 — Assemble and sketch

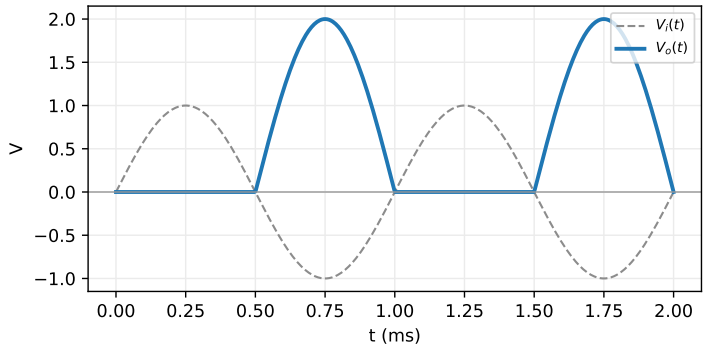
Put the valid states together:

$$V_o = \begin{cases} -2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



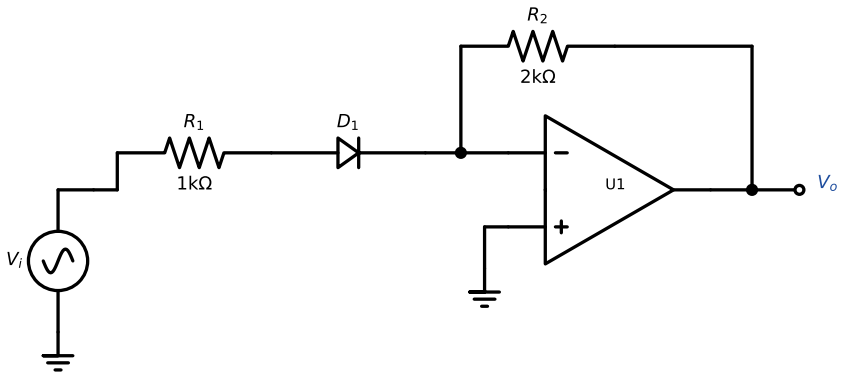
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Solution 3B.5 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input; V_a = node joining D_1, R_1.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- \,(=0): && \frac{0 - V_o}{R_2} = 0 \\ &\text{KCL at } V_a: && \frac{V_a - V_i}{R_1} = 0 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= 0 \\ V_{D_1} &= V_a - V_- = V_i \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$. So this state holds for $\boxed{V_i < 0.7}$, where $V_o = 0$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_a - V_- = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- \,(=0): && \frac{0 - V_o}{R_2} - I_{D_1} = 0 \\ &\text{KCL at } V_a: && \frac{V_a - V_i}{R_1} + I_{D_1} = 0 \\ &\text{D1 ON}: && V_a - 0 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -2V_i + 1.4$$
$$I_{D_1} = V_i - 0.7 \text{ mA}$$

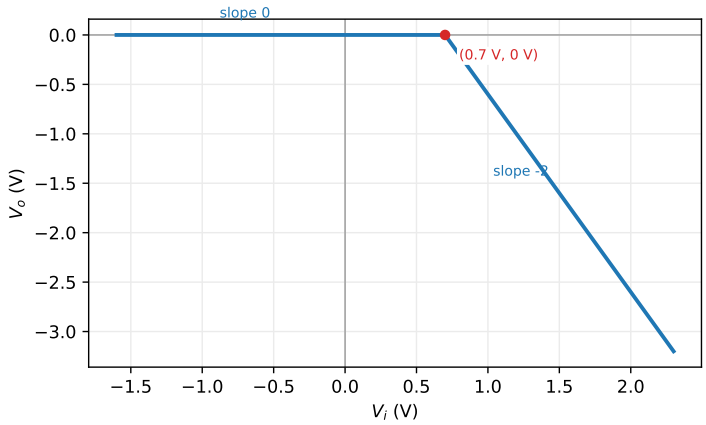
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.7\,\text{V}$. So this state holds for $\boxed{V_i > 0.7}$, where $V_o = -2V_i + 1.4$.

Step 4 — Assemble and sketch

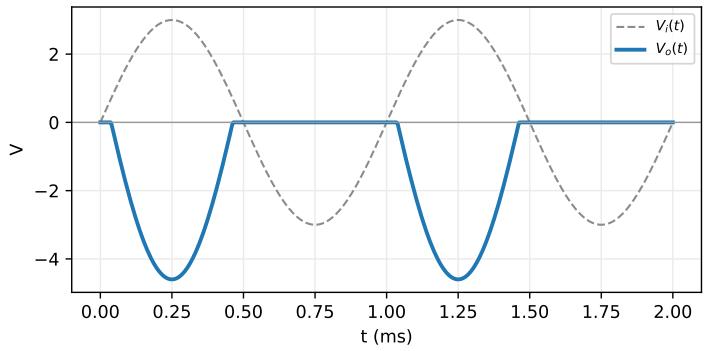
Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0.7 \\ -2V_i + 1.4 & V_i > 0.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.7\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



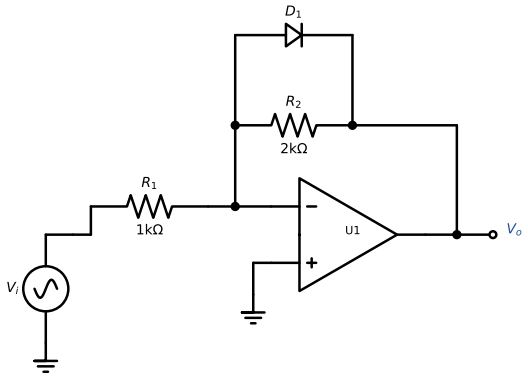
For $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0 & V_i < 0.7 \\ -2V_i + 1.4 & V_i > 0.7 \end{cases}$$

Solution 3B.6 **Easy**



Step 1 — Model and method

Use the ideal-diode model (tool T1): an OFF diode is an open circuit and needs $V_D < 0$; an ON diode is a short ($V_D = 0$) and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -2V_i$$
$$V_{D_1} = V_- - V_o = 2V_i$$

Check the assumptions: $V_{D_1} < 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = -2V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0 V source ($V_- - V_o = 0$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad 0 - V_o = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$

$$I_{D_1} = V_i \text{ mA}$$

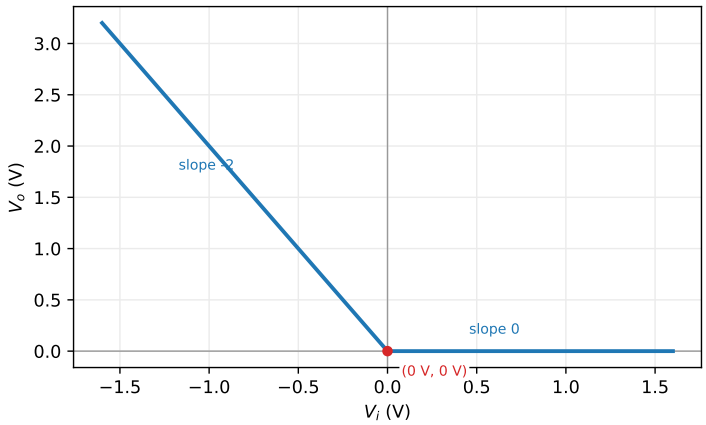
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0$.

Step 4 — Assemble and sketch

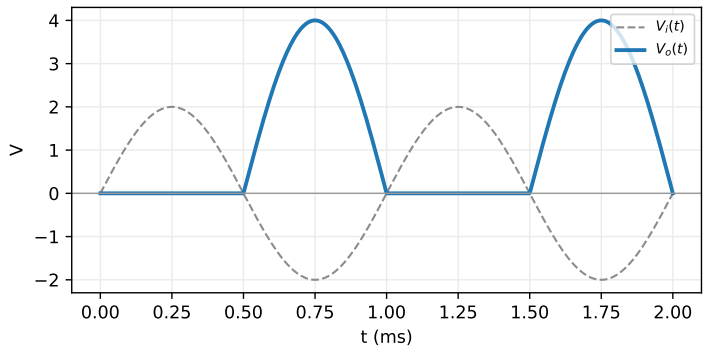
Put the valid states together:

$$V_o = \begin{cases} -2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



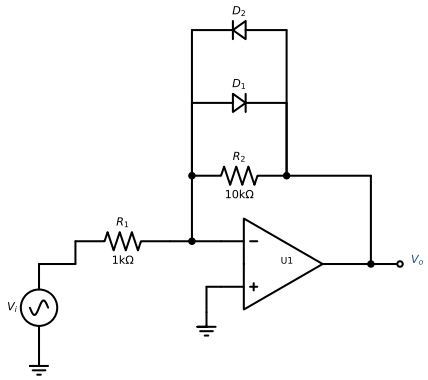
For $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Solution 3B.7 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_o - V_- = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} - I_{D_2} = 0 \\ &\text{D2 ON}: && V_o - 0 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= 0.7 \\ V_{D_1} = V_- - V_o &= -0.7 \\ I_{D_2} &= -V_i - 0.07 \text{ mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow$ true for every V_i; $I_{D_2} > 0 \Rightarrow V_i < -0.07\,\text{V}$. So this state holds for $\boxed{V_i < -0.07}$, where $V_o = 0.7$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -10V_i \\ V_{D_1} &= V_- - V_o = 10V_i \\ V_{D_2} &= V_o - V_- = -10V_i \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.07\,\text{V}$. So this state holds for $\boxed{-0.07 < V_i < 0.07}$, where $V_o = -10V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_- - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_- (= 0): && \frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} + I_{D_1} = 0 \\ &\text{D1 ON}: && 0 - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -0.7 \\ I_{D_1} &= V_i - 0.07\ \text{mA} \\ V_{D_2} &= V_o - V_- = -0.7 \end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0.07}$, where $V_o = -0.7$.

Step 5 — The other combinations

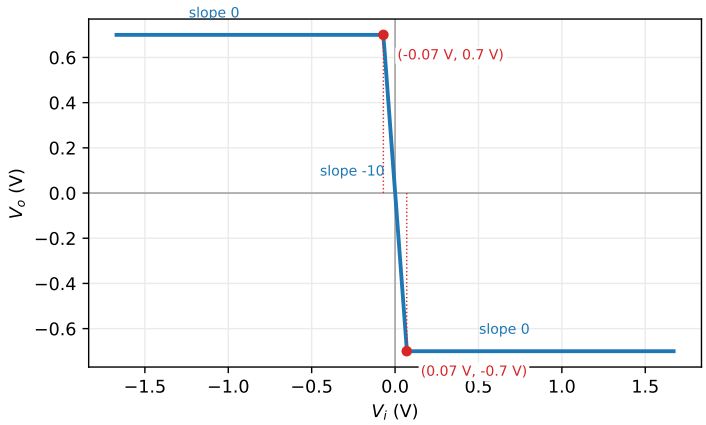
- D_1 ON, D_2 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

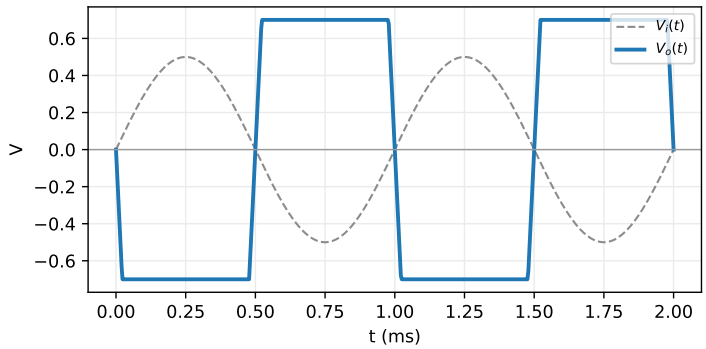
Put the valid states together:

$$V_o = \begin{cases} 0.7 & V_i < -0.07 \\ -10V_i & -0.07 < V_i < 0.07 \\ -0.7 & V_i > 0.07 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -0.07\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$; at $V_i = 0.07\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



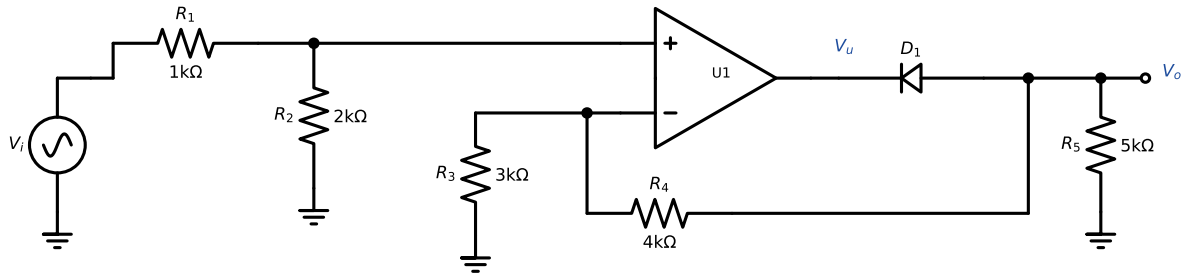
For $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0.7 & V_i < -0.07 \\ -10V_i & -0.07 < V_i < 0.07 \\ -0.7 & V_i > 0.07 \end{cases}$$

Solution 3B.8 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_+ = the op-amp's + input; V_- = the op-amp's − input.

Step 2 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_o - V_u = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_+ : && \frac{V_+ - V_i}{R_1} + \frac{V_+}{R_2} = 0 \\
&\text{KCL at } V_- (= V_+) : && \frac{V_+}{R_3} + \frac{V_+ - V_o}{R_4} = 0 \\
&\text{KCL at } V_o : && \frac{V_o - V_+}{R_4} + \frac{V_o}{R_5} + I_{D_1} = 0 \\
&\text{D1 ON} : && V_o - V_u = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 1.556 V_i \\
I_{D_1} &= -0.5333 V_i \text{ mA}
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = 1.556 V_i$.

Step 3 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. With this assumption U1 has no feedback path from its output back to its − input: it runs *open-loop*, $V_u = A(V_+ - V_-)$ with $A \to \infty$, so its output is pushed to $\pm\infty$ (in practice, a supply rail) unless $V_+ = V_-$ exactly.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_+ : && \frac{V_+ - V_i}{R_1} + \frac{V_+}{R_2} = 0 \\
&\text{KCL at } V_- : && \frac{V_-}{R_3} + \frac{V_- - V_o}{R_4} = 0
\end{aligned}$$

$$\text{KCL at } V_o: \qquad \frac{V_o - V_-}{R_4} + \frac{V_o}{R_5} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$V_{D_1} = V_o - V_u \approx A\,(-0.6667 V_i)$$

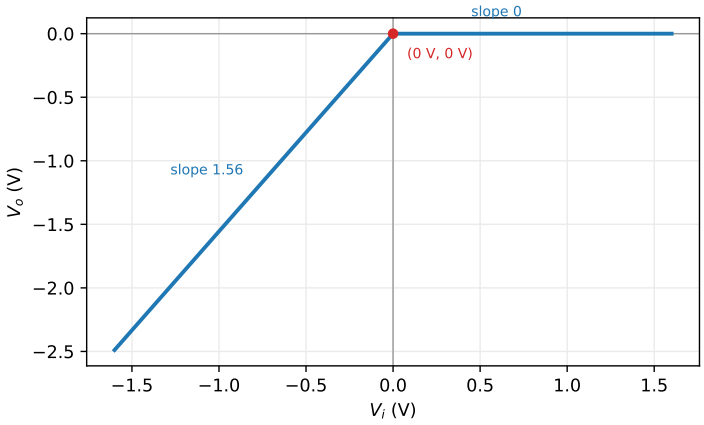
Check the assumptions: $V_{D_1} < 0.7$; since $A \to \infty$ this needs $-0.6667 V_i < 0 \Rightarrow V_i > 0\,\text{V}$. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0$.

Step 4 — Assemble and sketch

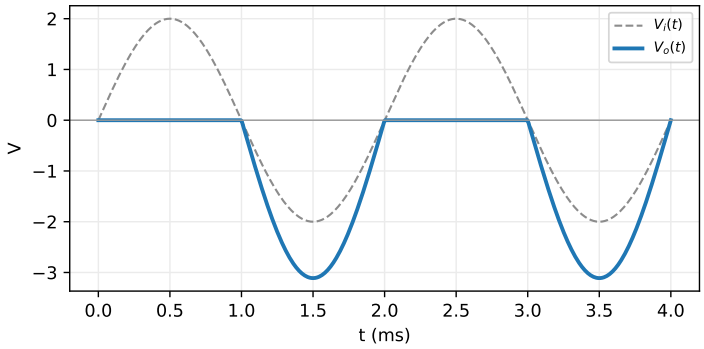
Put the valid states together:

$$V_o = \begin{cases} 1.556 V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



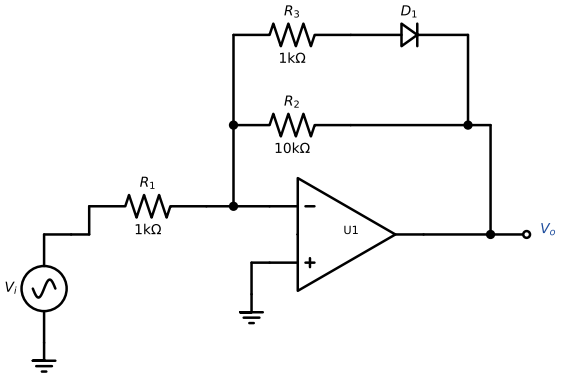
For $V_i(t) = 2\sin(2\pi \cdot 500\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 1.556V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Solution 3B.9 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input; V_a = node joining D_1, R_3.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \quad \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} + \frac{0-V_a}{R_3} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a}{R_3} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -10V_i$$
$$V_{D_1} = V_a - V_o = 10V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.07\,\text{V}$. So this state holds for $\boxed{V_i < 0.07}$, where $V_o = -10V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_a - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \quad \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} + \frac{0-V_a}{R_3} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a}{R_3} + I_{D_1} = 0$$

$$\text{D1 ON}: \quad V_a - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -0.9091V_i - 0.6364$$
$$I_{D_1} = 0.9091V_i - 0.06364 \text{ mA}$$

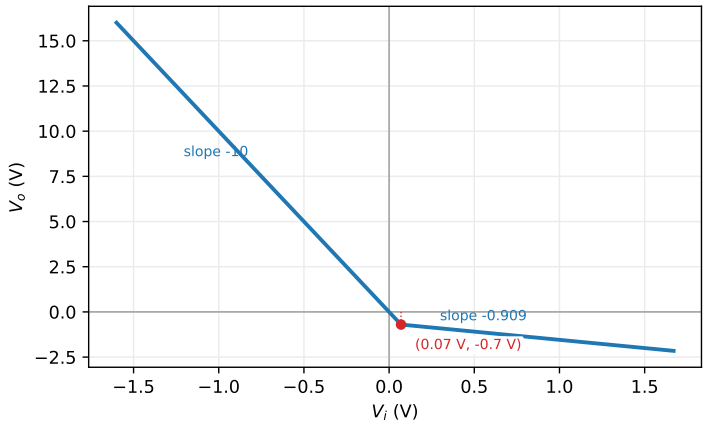
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.07\,\text{V}$. So this state holds for $\boxed{V_i > 0.07}$, where $V_o = -0.9091V_i - 0.6364$.

Step 4 — Assemble and sketch

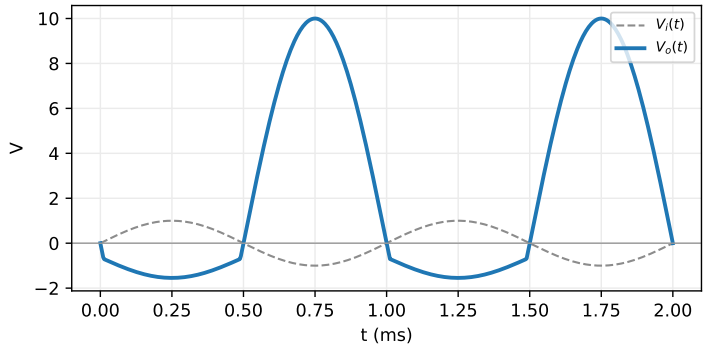
Put the valid states together:

$$V_o = \begin{cases} -10V_i & V_i < 0.07 \\ -0.9091V_i - 0.6364 & V_i > 0.07 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.07\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



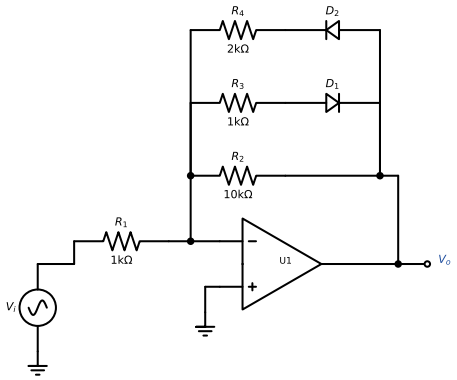
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -10V_i & V_i < 0.07 \\ -0.9091V_i - 0.6364 & V_i > 0.07 \end{cases}$$

Solution 3B.10 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input; V_a = node joining D_1, R_3; V_b = node joining D_2, R_4.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_o - V_b = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1} + \frac{0-V_o}{R_2} + \frac{0-V_a}{R_3} + \frac{0-V_b}{R_4} = 0 \\
&\text{KCL at } V_a: && \frac{V_a}{R_3} = 0 \\
&\text{KCL at } V_b: && \frac{V_b}{R_4} - I_{D_2} = 0 \\
&\text{D2 ON}: && V_o - V_b = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -1.667V_i + 0.5833 \\
V_{D_1} = V_a - V_o &= 1.667V_i - 0.5833 \\
I_{D_2} &= -0.8333V_i - 0.05833\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.77\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -0.07\,\text{V}$. So this state holds for $\boxed{V_i < -0.07}$, where $V_o = -1.667V_i + 0.5833$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1}+\frac{0-V_o}{R_2}+\frac{0-V_a}{R_3}+\frac{0-V_b}{R_4}=0\\
&\text{KCL at } V_a: && \frac{V_a}{R_3}=0\\
&\text{KCL at } V_b: && \frac{V_b}{R_4}=0
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -10V_i\\
V_{D_1} = V_a - V_o &= 10V_i\\
V_{D_2} = V_o - V_b &= -10V_i
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.07\,\text{V}$. So this state holds for $\boxed{-0.07 < V_i < 0.07}$, where $V_o = -10V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_a - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1}+\frac{0-V_o}{R_2}+\frac{0-V_a}{R_3}+\frac{0-V_b}{R_4}=0\\
&\text{KCL at } V_a: && \frac{V_a}{R_3}+I_{D_1}=0\\
&\text{KCL at } V_b: && \frac{V_b}{R_4}=0\\
&\text{D1 ON}: && V_a - V_o = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -0.9091V_i - 0.6364\\
I_{D_1} &= 0.9091V_i - 0.06364\ \text{mA}\\
V_{D_2} = V_o - V_b &= -0.9091V_i - 0.6364
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -1.47\,\text{V}$. So this state holds for $\boxed{V_i > 0.07}$, where $V_o = -0.9091V_i - 0.6364$.

Step 5 — The other combinations

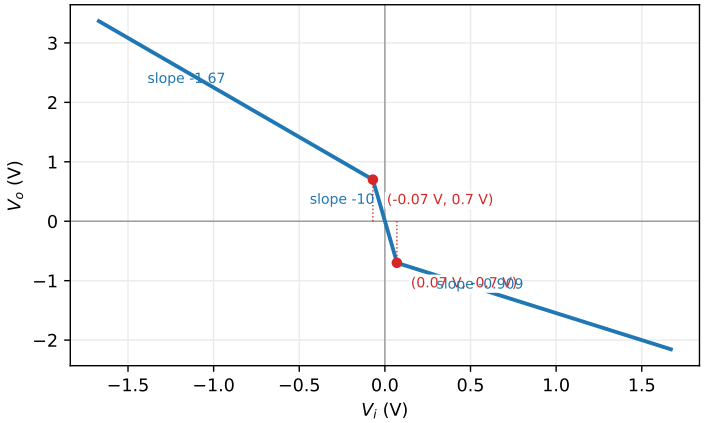
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 0.77$; $I_{D_2} > 0$ needs $V_i < -1.47$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

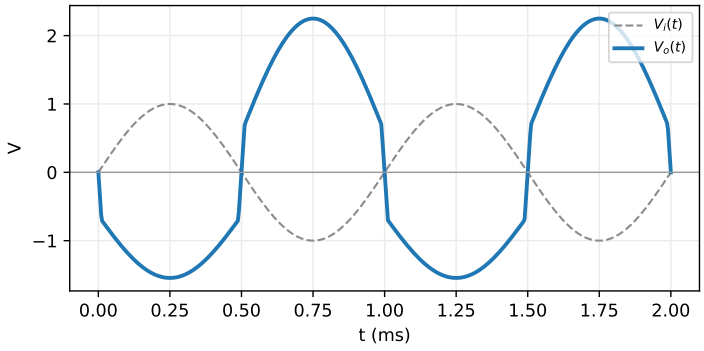
Put the valid states together:

$$V_o = \begin{cases} -1.667V_i + 0.5833 & V_i < -0.07 \\ -10V_i & -0.07 < V_i < 0.07 \\ -0.9091V_i - 0.6364 & V_i > 0.07 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -0.07\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$; at $V_i = 0.07\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



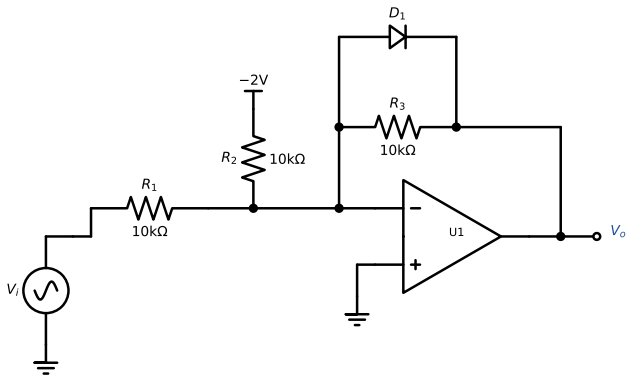
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -1.667V_i + 0.5833 & V_i < -0.07 \\ -10V_i & -0.07 < V_i < 0.07 \\ -0.9091V_i - 0.6364 & V_i > 0.07 \end{cases}$$

Solution 3B.11 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0-V_i}{R_1} + \frac{0-V_o}{R_3} + \frac{0--2}{R_2} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -V_i + 2$$
$$V_{D_1} = V_- - V_o = V_i - 2$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.7\,\text{V}$. So this state holds for $\boxed{V_i < 2.7}$, where $V_o = -V_i + 2$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_- - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \qquad \frac{0-V_i}{R_1} + \frac{0-V_o}{R_3} + \frac{0--2}{R_2} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad 0 - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -0.7$$

$$I_{D_1} = 0.1V_i - 0.27 \text{ mA}$$

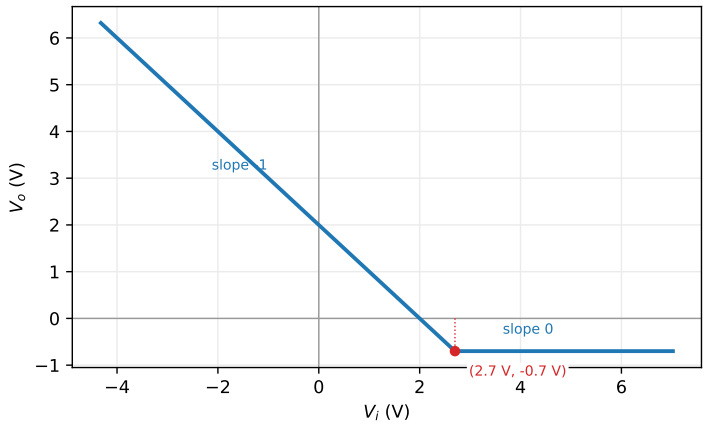
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2.7\,\text{V}$. So this state holds for $\boxed{V_i > 2.7}$, where $V_o = -0.7$.

Step 4 — Assemble and sketch

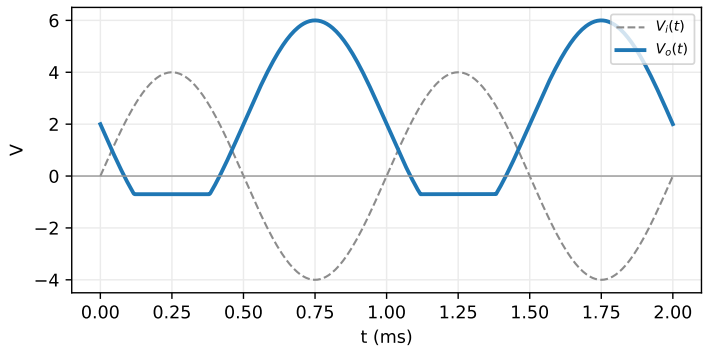
Put the valid states together:

$$V_o = \begin{cases} -V_i + 2 & V_i < 2.7 \\ -0.7 & V_i > 2.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 2.7\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



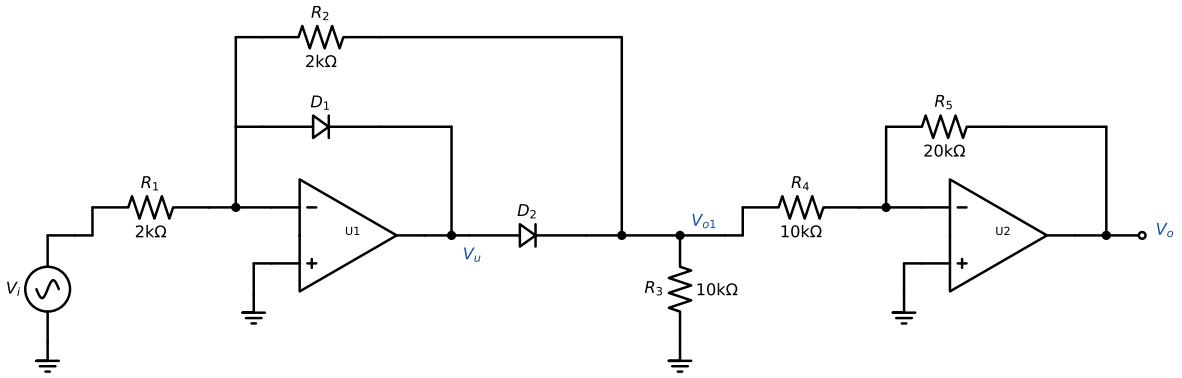
For $V_i(t) = 4\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -V_i + 2 & V_i < 2.7 \\ -0.7 & V_i > 2.7 \end{cases}$$

Solution 3B.12 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_{-1} = node joining $D_1, R_1, R_2, theop-ampoutput$; V_{-2} = node joining $R_4, R_5, theop-ampoutput$; V_{o1} = node joining D_2, R_2, R_3, R_4; V_{u1} = node joining $D_1, D_2, theop-ampoutput$.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_{u1} - V_{o1} = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} = 0 \\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_4} + \frac{0-V_o}{R_5} = 0 \\
&\text{KCL at } V_{o1}: && \frac{V_{o1}}{R_2} + \frac{V_{o1}}{R_3} + \frac{V_{o1}}{R_4} - I_{D_2} = 0 \\
&\text{D2 ON}: && V_{u1} - V_{o1} = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 2V_i \\
V_{D_1} = V_{-1} - V_{u1} &= V_i - 0.7 \\
I_{D_2} &= -0.7V_i \text{ mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 1.4\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = 2V_i$.

Step 3 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_{-1} - V_{u1} = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1}+\frac{0-V_{o1}}{R_2}+I_{D_1}=0\\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_4}+\frac{0-V_o}{R_5}=0\\
&\text{KCL at } V_{o1}: && \frac{V_{o1}}{R_2}+\frac{V_{o1}}{R_3}+\frac{V_{o1}}{R_4}=0\\
&\text{D1 ON}: && 0-V_{u1}=0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0\\
I_{D_1} &= 0.5V_i \text{ mA}\\
V_{D_2} = V_{u1} - V_{o1} &= -0.7
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0$.

Step 4 — The other combinations

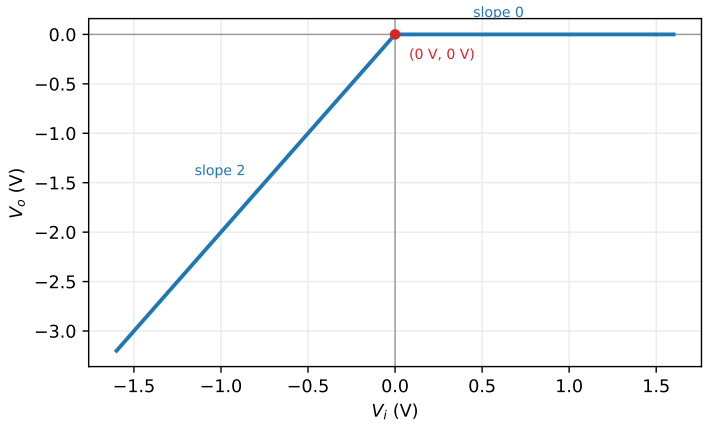
- D_1 OFF, D_2 OFF: $V_{D_1} < 0.7$ needs $V_i < 0$; $V_{D_2} < 0.7$ needs $V_i > 0$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 1.4$; $I_{D_2} > 0$ is never true — these cannot all hold, so this state never occurs.

Step 5 — Assemble and sketch

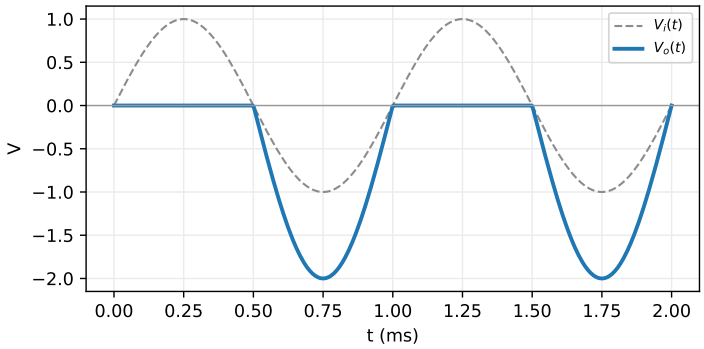
Put the valid states together:

$$V_o = \begin{cases} 2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



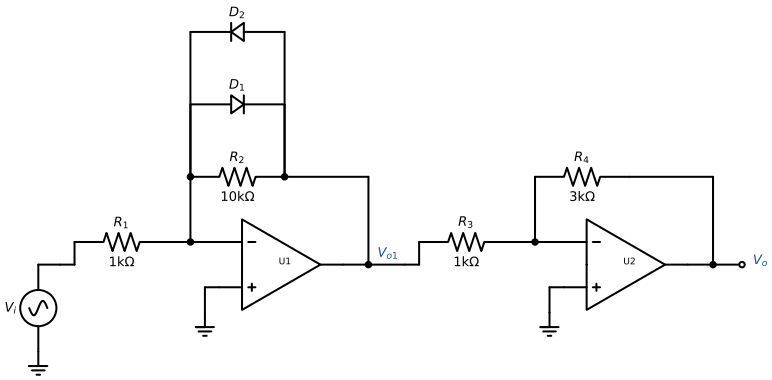
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 2V_i & V_i < 0 \\ 0 & V_i > 0 \end{cases}$$

Solution 3B.13 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_{-1} = node joining $D_1, D_2, R_1, R_2, \mathit{theop-ampoutput}$; V_{-2} = node joining $R_3, R_4, \mathit{theop-ampoutput}$; V_{o1} = node joining $D_1, D_2, R_2, R_3, \mathit{theop-ampoutput}$.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_{o1} - V_{-1} = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} - I_{D_2} = 0 \\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_3} + \frac{0-V_o}{R_4} = 0 \\
&\text{D2 ON}: && V_{o1} - 0 = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -2.1 \\
V_{D_1} = V_{-1} - V_{o1} &= -0.7 \\
I_{D_2} &= -V_i - 0.07\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow$ true for every V_i; $I_{D_2} > 0 \Rightarrow V_i < -0.07\,\text{V}$. So this state holds for $\boxed{V_i < -0.07}$, where $V_o = -2.1$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_{-1}\,(=0): \qquad \frac{0-V_i}{R_1}+\frac{0-V_{o1}}{R_2}=0$$

$$\text{KCL at } V_{-2}\,(=0): \qquad \frac{0-V_{o1}}{R_3}+\frac{0-V_o}{R_4}=0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 30V_i$$
$$V_{D_1} = V_{-1} - V_{o1} = 10V_i$$
$$V_{D_2} = V_{o1} - V_{-1} = -10V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.07\,\text{V}$. So this state holds for $\boxed{-0.07 < V_i < 0.07}$, where $V_o = 30V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_{-1} - V_{o1} = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_{-1}\,(=0): \qquad \frac{0-V_i}{R_1}+\frac{0-V_{o1}}{R_2}+I_{D_1}=0$$

$$\text{KCL at } V_{-2}\,(=0): \qquad \frac{0-V_{o1}}{R_3}+\frac{0-V_o}{R_4}=0$$

$$\text{D1 ON}: \qquad 0-V_{o1}=0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 2.1$$
$$I_{D_1} = V_i - 0.07\ \text{mA}$$
$$V_{D_2} = V_{o1} - V_{-1} = -0.7$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.07\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0.07}$, where $V_o = 2.1$.

Step 5 — The other combinations

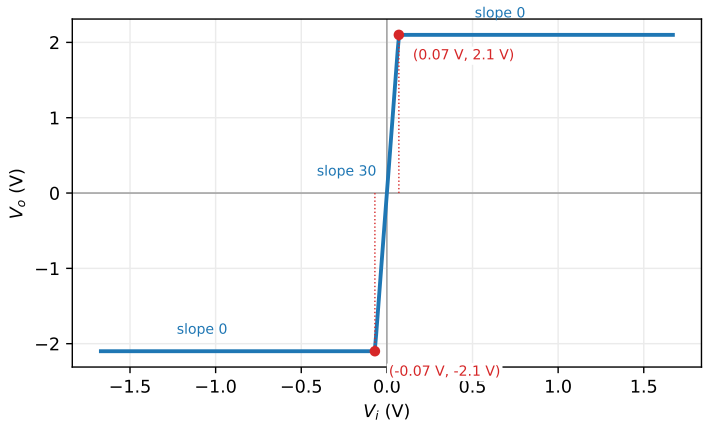
- D_1 ON, D_2 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

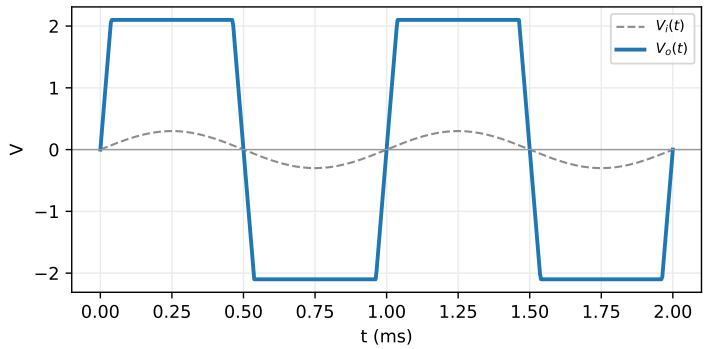
Put the valid states together:

$$V_o = \begin{cases} -2.1 & V_i < -0.07 \\ 30V_i & -0.07 < V_i < 0.07 \\ 2.1 & V_i > 0.07 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -0.07\,\text{V}$ both pieces give $V_o = -2.1\,\text{V}$; at $V_i = 0.07\,\text{V}$ both pieces give $V_o = 2.1\,\text{V}$. ✓



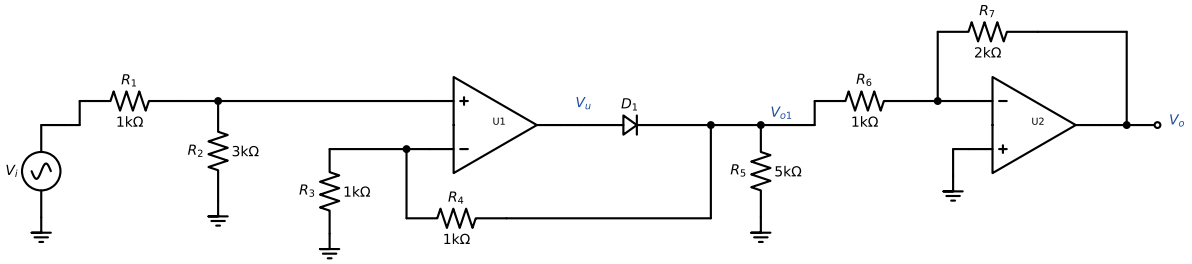
For $V_i(t) = 0.3\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -2.1 & V_i < -0.07 \\ 30V_i & -0.07 < V_i < 0.07 \\ 2.1 & V_i > 0.07 \end{cases}$$

Solution 3B.14 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_{+1} = node joining $R_1, R_2, \mathit{theop-ampoutput}$; V_{-1} = node joining $R_3, R_4, \mathit{theop-ampoutput}$; V_{-2} = node joining $R_6, R_7, \mathit{theop-ampoutput}$; V_{o1} = node joining D_1, R_4, R_5, R_6; V_{u1} = node joining $D_1, \mathit{theop-ampoutput}$.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. With this assumption U1 has no feedback path from its output back to its − input: it runs *open-loop*, $V_{u1} = A(V_+ - V_-)$ with $A \to \infty$, so its output is pushed to $\pm\infty$ (in practice, a supply rail) unless $V_+ = V_-$ exactly. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_{+1}: \quad \frac{V_{+1} - V_i}{R_1} + \frac{V_{+1}}{R_2} = 0$$

$$\text{KCL at } V_{-1}: \quad \frac{V_{-1}}{R_3} + \frac{V_{-1} - V_{o1}}{R_4} = 0$$

$$\text{KCL at } V_{-2}\,(=0): \quad \frac{0 - V_{o1}}{R_6} + \frac{0 - V_o}{R_7} = 0$$

$$\text{KCL at } V_{o1}: \quad \frac{V_{o1} - V_{-1}}{R_4} + \frac{V_{o1}}{R_5} + \frac{V_{o1}}{R_6} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$V_{D_1} = V_{u1} - V_{o1} \approx A\,(0.75 V_i)$$

Check the assumptions: $V_{D_1} < 0.7$; since $A \to \infty$ this needs $0.75 V_i < 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = 0$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_{u1} - V_{o1} = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{+1}: && \frac{V_{+1}-V_i}{R_1}+\frac{V_{+1}}{R_2}=0\\
&\text{KCL at } V_{-1}\,(=V_{+1}): && \frac{V_{+1}}{R_3}+\frac{V_{+1}-V_{o1}}{R_4}=0\\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_6}+\frac{0-V_o}{R_7}=0\\
&\text{KCL at } V_{o1}: && \frac{V_{o1}-V_{+1}}{R_4}+\frac{V_{o1}}{R_5}+\frac{V_{o1}}{R_6}-I_{D_1}=0\\
&\text{D1 ON}: && V_{u1}-V_{o1}=0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -3V_i\\
I_{D_1} &= 2.55V_i \text{ mA}
\end{aligned}$$

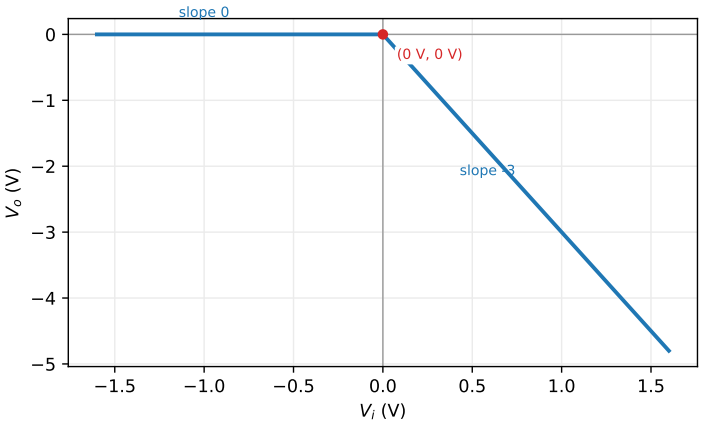
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$. So this state holds for $\boxed{V_i > 0}$, where $V_o = -3V_i$.

Step 4 — Assemble and sketch

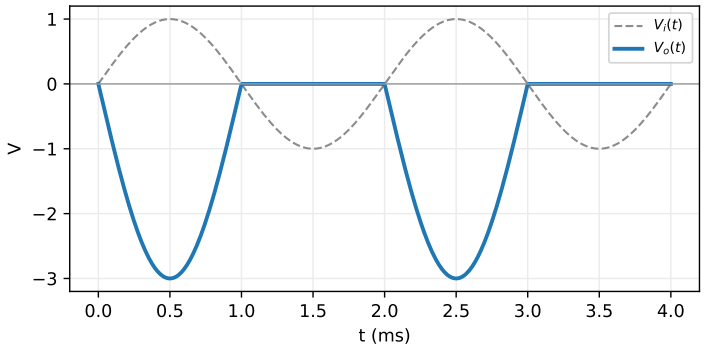
Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0 \\ -3V_i & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



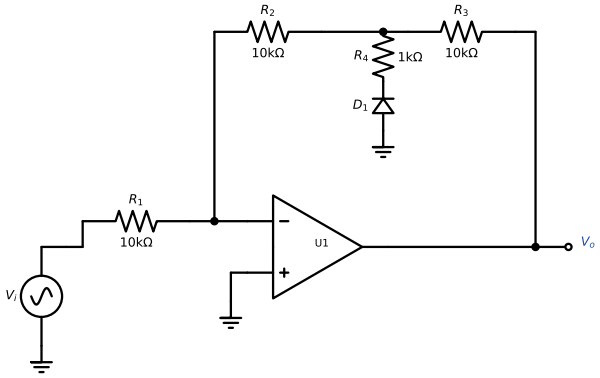
For $V_i(t) = 1\sin(2\pi \cdot 500\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0 & V_i < 0 \\ -3V_i & V_i > 0 \end{cases}$$

Solution 3B.15 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's − input; V_a = node joining D_1, R_4; V_b = node joining R_2, R_3, R_4.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \quad \frac{0 - V_i}{R_1} + \frac{0 - V_b}{R_2} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a - V_b}{R_4} = 0$$

$$\text{KCL at } V_b: \quad \frac{V_b}{R_2} + \frac{V_b - V_o}{R_3} + \frac{V_b - V_a}{R_4} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -2V_i$$
$$V_{D_1} = 0 - V_a = V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$. So this state holds for $\boxed{V_i < 0.7}$, where $V_o = -2V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($0 - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_- (=0): \quad \frac{0 - V_i}{R_1} + \frac{0 - V_b}{R_2} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a - V_b}{R_4} - I_{D_1} = 0$$

KCL at V_b :	$\dfrac{V_b}{R_2} + \dfrac{V_b - V_o}{R_3} + \dfrac{V_b - V_a}{R_4} = 0$
D1 ON :	$0 - V_a = 0.7$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -12V_i + 7$$
$$I_{D_1} = V_i - 0.7 \text{ mA}$$

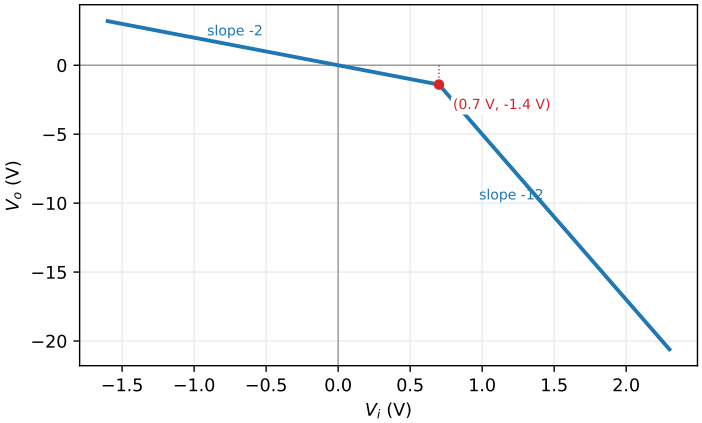
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.7\,\text{V}$. So this state holds for $\boxed{V_i > 0.7}$, where $V_o = -12V_i + 7$.

Step 4 — Assemble and sketch

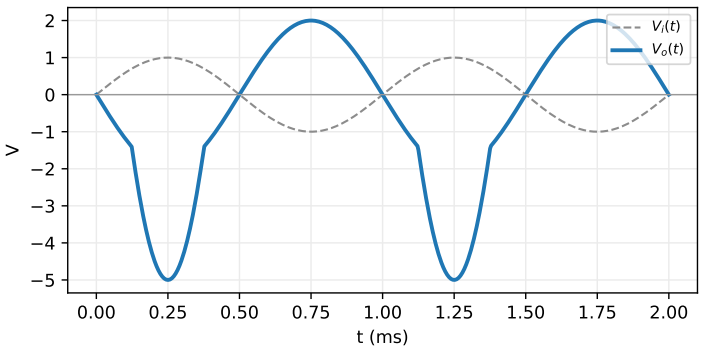
Put the valid states together:

$$V_o = \begin{cases} -2V_i & V_i < 0.7 \\ -12V_i + 7 & V_i > 0.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.7\,\text{V}$ both pieces give $V_o = -1.4\,\text{V}$. ✓



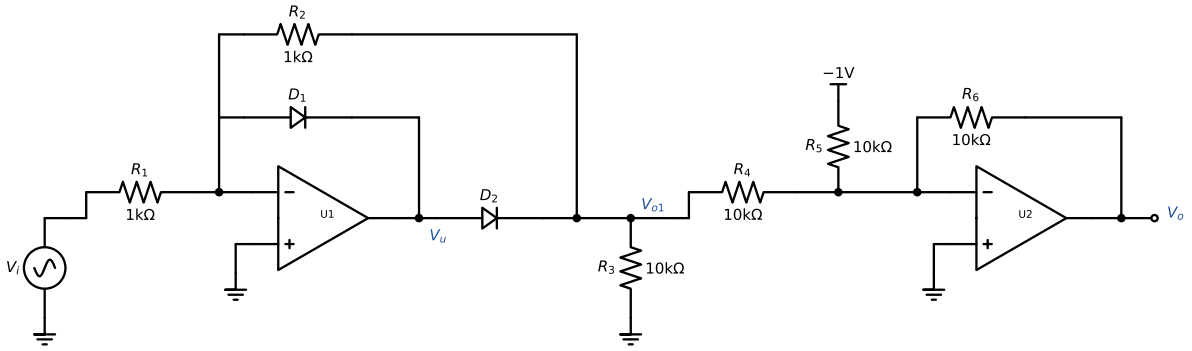
For $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -2V_i & V_i < 0.7 \\ -12V_i + 7 & V_i > 0.7 \end{cases}$$

Solution 3B.16 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_{-1} = node joining $D_1, R_1, R_2, \mathit{theop-ampoutput}$; V_{-2} = node joining $R_4, R_5, R_6, \mathit{theop-ampoutput}$; V_{o1} = node joining D_2, R_2, R_3, R_4; V_{u1} = node joining $D_1, D_2, \mathit{theop-ampoutput}$.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_{u1} - V_{o1} = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} = 0\\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_4} + \frac{0-V_o}{R_6} + \frac{0--1}{R_5} = 0\\
&\text{KCL at } V_{o1}: && \frac{V_{o1}}{R_2} + \frac{V_{o1}}{R_3} + \frac{V_{o1}}{R_4} - I_{D_2} = 0\\
&\text{D2 ON}: && V_{u1} - V_{o1} = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= V_i + 1\\
V_{D_1} = V_{-1} - V_{u1} &= V_i - 0.7\\
I_{D_2} &= -1.2V_i \text{ mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 1.4\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = V_i + 1$.

Step 3 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_{-1} - V_{u1} = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1}+\frac{0-V_{o1}}{R_2}+I_{D_1}=0\\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_4}+\frac{0-V_o}{R_6}+\frac{0--1}{R_5}=0\\
&\text{KCL at } V_{o1}: && \frac{V_{o1}}{R_2}+\frac{V_{o1}}{R_3}+\frac{V_{o1}}{R_4}=0\\
&\text{D1 ON}: && 0-V_{u1}=0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 1\\
I_{D_1} &= V_i \text{ mA}\\
V_{D_2} = V_{u1} - V_{o1} &= -0.7
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0}$, where $V_o = 1$.

Step 4 — The other combinations

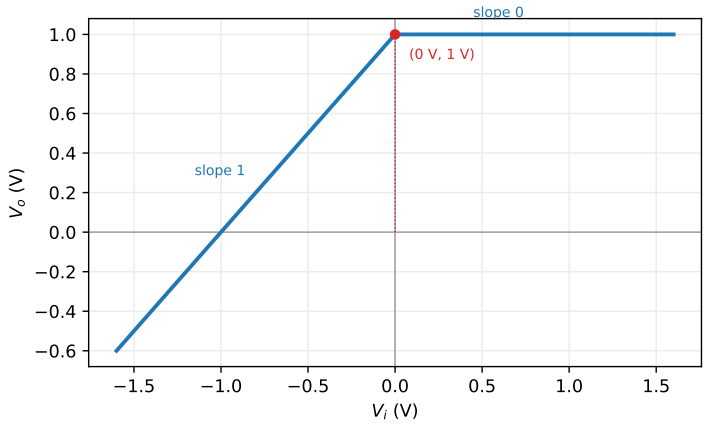
- D_1 OFF, D_2 OFF: $V_{D_1} < 0.7$ needs $V_i < 0$; $V_{D_2} < 0.7$ needs $V_i > 0$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 1.4$; $I_{D_2} > 0$ is never true — these cannot all hold, so this state never occurs.

Step 5 — Assemble and sketch

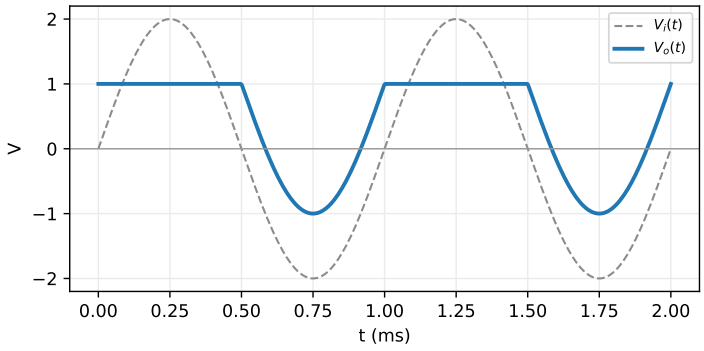
Put the valid states together:

$$V_o = \begin{cases} V_i + 1 & V_i < 0 \\ 1 & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 1\,\text{V}$. ✓



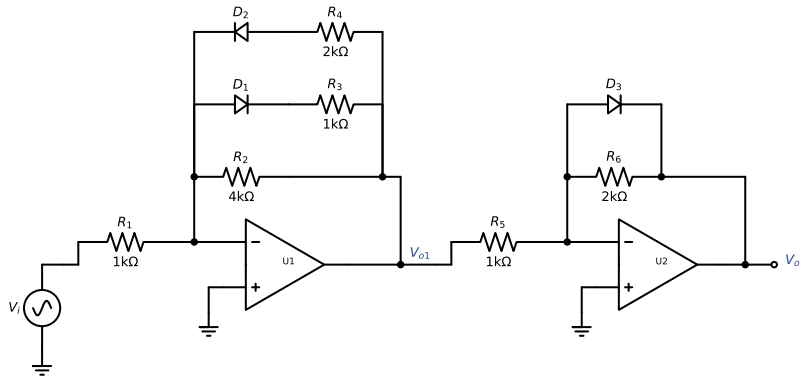
For $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} V_i + 1 & V_i < 0 \\ 1 & V_i > 0 \end{cases}$$

Solution 3B.17 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 3 diodes there are 8 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_{-1} = node joining $D_1, D_2, R_1, R_2, \mathit{theop-ampoutput}$; V_{-2} = node joining $D_3, R_5, R_6, \mathit{theop-ampoutput}$; V_a = node joining D_1, R_3; V_b = node joining D_2, R_4; V_{o1} = node joining $R_2, R_3, R_4, R_5, \mathit{theop-ampoutput}$.

Step 2 — Assume D_1 OFF, D_2 ON, D_3 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_b - V_{-1} = 0.7$) carrying an unknown current I_{D_2} from anode to cathode; D_3 ON: a 0.7 V source ($V_{-2} - V_o = 0.7$) carrying an unknown current I_{D_3} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} - I_{D_2} = 0 \\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_5} + \frac{0-V_o}{R_6} + I_{D_3} = 0 \\
&\text{KCL at } V_a: && \frac{V_a - V_{o1}}{R_3} = 0 \\
&\text{KCL at } V_b: && \frac{V_b - V_{o1}}{R_4} + I_{D_2} = 0 \\
&\text{D2 ON}: && V_b - 0 = 0.7 \\
&\text{D3 ON}: && 0 - V_o = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -0.7 \\
V_{D_1} = V_{-1} - V_a &= 1.333V_i - 0.4667 \\
I_{D_2} &= -0.6667V_i - 0.1167\ \text{mA} \\
I_{D_3} &= -1.333V_i + 0.1167\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.875\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -0.175\,\text{V}$; $I_{D_3} > 0 \Rightarrow V_i < 0.0875\,\text{V}$. So this state holds for $\boxed{V_i < -0.175}$, where $V_o = -0.7$.

Step 3 — Assume D_1 OFF, D_2 OFF, D_3 ON

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit; D_3 ON: a 0.7 V source ($V_{-2} - V_o = 0.7$) carrying an unknown current I_{D_3} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} = 0 \\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_5} + \frac{0-V_o}{R_6} + I_{D_3} = 0 \\
&\text{KCL at } V_a: && \frac{V_a - V_{o1}}{R_3} = 0 \\
&\text{KCL at } V_b: && \frac{V_b - V_{o1}}{R_4} = 0 \\
&\text{D3 ON}: && 0 - V_o = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -0.7 \\
V_{D_1} = V_{-1} - V_a &= 4V_i \\
V_{D_2} = V_b - V_{-1} &= -4V_i \\
I_{D_3} &= -4V_i - 0.35\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.175\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.175\,\text{V}$; $I_{D_3} > 0 \Rightarrow V_i < -0.0875\,\text{V}$. So this state holds for $\boxed{-0.175 < V_i < -0.0875}$, where $V_o = -0.7$.

Step 4 — Assume D_1 OFF, D_2 OFF, D_3 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit; D_3 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_{-1}\,(=0): && \frac{0-V_i}{R_1} + \frac{0-V_{o1}}{R_2} = 0 \\
&\text{KCL at } V_{-2}\,(=0): && \frac{0-V_{o1}}{R_5} + \frac{0-V_o}{R_6} = 0 \\
&\text{KCL at } V_a: && \frac{V_a - V_{o1}}{R_3} = 0 \\
&\text{KCL at } V_b: && \frac{V_b - V_{o1}}{R_4} = 0
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 8V_i$$

$$V_{D_1} = V_{-1} - V_a = 4V_i$$
$$V_{D_2} = V_b - V_{-1} = -4V_i$$
$$V_{D_3} = V_{-2} - V_o = -8V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.175\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.175\,\text{V}$; $V_{D_3} < 0.7 \Rightarrow V_i > -0.0875\,\text{V}$. So this state holds for $\boxed{-0.0875 < V_i < 0.175}$, where $V_o = 8V_i$.

Step 5 — Assume D_1 ON, D_2 OFF, D_3 OFF

Replace D_1 ON: a 0.7 V source ($V_{-1} - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit; D_3 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs. U2 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_{-1}\,(=0): \quad \frac{0 - V_i}{R_1} + \frac{0 - V_{o1}}{R_2} + I_{D_1} = 0$$
$$\text{KCL at } V_{-2}\,(=0): \quad \frac{0 - V_{o1}}{R_5} + \frac{0 - V_o}{R_6} = 0$$
$$\text{KCL at } V_a: \quad \frac{V_a - V_{o1}}{R_3} - I_{D_1} = 0$$
$$\text{KCL at } V_b: \quad \frac{V_b - V_{o1}}{R_4} = 0$$
$$\text{D1 ON}: \quad 0 - V_a = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 1.6V_i + 1.12$$
$$I_{D_1} = 0.8V_i - 0.14\ \text{mA}$$
$$V_{D_2} = V_b - V_{-1} = -0.8V_i - 0.56$$
$$V_{D_3} = V_{-2} - V_o = -1.6V_i - 1.12$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.175\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -1.575\,\text{V}$; $V_{D_3} < 0.7 \Rightarrow V_i > -1.137\,\text{V}$. So this state holds for $\boxed{V_i > 0.175}$, where $V_o = 1.6V_i + 1.12$.

Step 6 — The other combinations

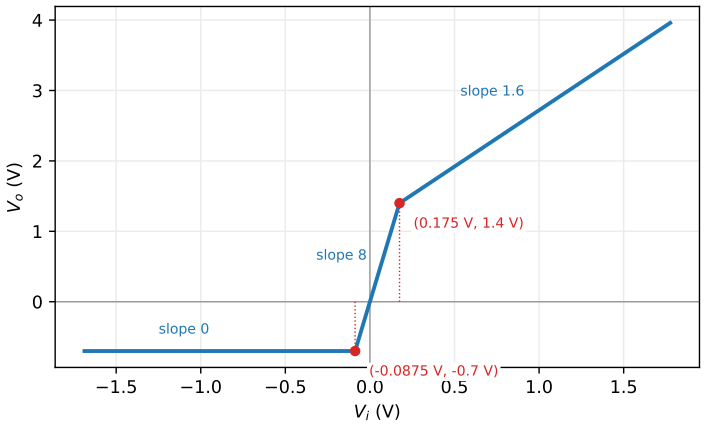
- D_1 OFF, D_2 ON, D_3 OFF: $V_{D_1} < 0.7$ needs $V_i < 0.875$; $I_{D_2} > 0$ needs $V_i < -0.175$; $V_{D_3} < 0.7$ needs $V_i > 0.0875$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 OFF, D_3 ON: $I_{D_1} > 0$ needs $V_i > 0.175$; $V_{D_2} < 0.7$ needs $V_i > -1.575$; $I_{D_3} > 0$ needs $V_i < -1.137$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON, D_3 OFF: $I_{D_1} > 0$ needs $V_i > 0.875$; $I_{D_2} > 0$ needs $V_i < -1.575$; $V_{D_3} < 0.7$ needs $V_i > -0.9625$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON, D_3 ON: $I_{D_1} > 0$ needs $V_i > 0.875$; $I_{D_2} > 0$ needs $V_i < -1.575$; $I_{D_3} > 0$ needs $V_i < -0.9625$ — these cannot all hold, so this state never occurs.

Step 7 — Assemble and sketch

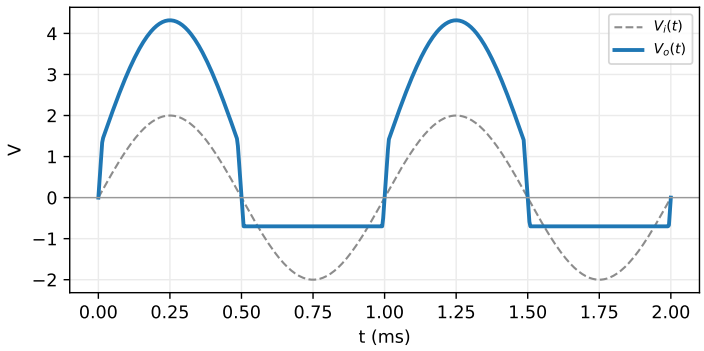
Put the valid states together (neighbouring states that give the *same* V_o are merged: a diode changes state there, but it does not affect V_o):

$$V_o = \begin{cases} -0.7 & V_i < -0.0875 \\ 8V_i & -0.0875 < V_i < 0.175 \\ 1.6V_i + 1.12 & V_i > 0.175 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -0.0875\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$; at $V_i = 0.175\,\text{V}$ both pieces give $V_o = 1.4\,\text{V}$. ✓



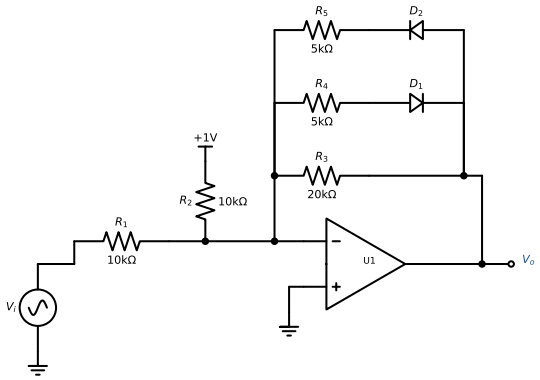
For $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -0.7 & V_i < -0.0875 \\ 8V_i & -0.0875 < V_i < 0.175 \\ 1.6V_i + 1.12 & V_i > 0.175 \end{cases}$$

Solution 3B.18 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Op-amps are ideal ($A \to \infty$). Whether an op-amp has negative feedback can depend on the diode states, so decide it separately for every combination. Node names used below: V_- = the op-amp's $-$ input; V_a = node joining D_1, R_4; V_b = node joining D_2, R_5.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_o - V_b = 0.7$) carrying an unknown current I_{D_2} from anode to cathode. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1} + \frac{0-V_o}{R_3} + \frac{0-V_a}{R_4} + \frac{0-V_b}{R_5} + \frac{0-1}{R_2} = 0 \\
&\text{KCL at } V_a: && \frac{V_a}{R_4} = 0 \\
&\text{KCL at } V_b: && \frac{V_b}{R_5} - I_{D_2} = 0 \\
&\text{D2 ON}: && V_o - V_b = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -0.4V_i + 0.16 \\
V_{D_1} = V_a - V_o &= 0.4V_i - 0.16 \\
I_{D_2} &= -0.08V_i - 0.108\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.15\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -1.35\,\text{V}$. So this state holds for $\boxed{V_i < -1.35}$, where $V_o = -0.4V_i + 0.16$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1}+\frac{0-V_o}{R_3}+\frac{0-V_a}{R_4}+\frac{0-V_b}{R_5}+\frac{0-1}{R_2}=0\\
&\text{KCL at } V_a: && \frac{V_a}{R_4}=0\\
&\text{KCL at } V_b: && \frac{V_b}{R_5}=0
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -2V_i - 2\\
V_{D_1} = V_a - V_o &= 2V_i + 2\\
V_{D_2} = V_o - V_b &= -2V_i - 2
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < -0.65\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -1.35\,\text{V}$. So this state holds for $\boxed{-1.35 < V_i < -0.65}$, where $V_o = -2V_i - 2$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_a - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit. U1 has negative feedback, so $V_- = V_+$ (virtual short) and no current flows into its inputs.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_- (=0): && \frac{0-V_i}{R_1}+\frac{0-V_o}{R_3}+\frac{0-V_a}{R_4}+\frac{0-V_b}{R_5}+\frac{0-1}{R_2}=0\\
&\text{KCL at } V_a: && \frac{V_a}{R_4}+I_{D_1}=0\\
&\text{KCL at } V_b: && \frac{V_b}{R_5}=0\\
&\text{D1 ON}: && V_a - V_o = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= -0.4V_i - 0.96\\
I_{D_1} &= 0.08V_i + 0.052\ \text{mA}\\
V_{D_2} = V_o - V_b &= -0.4V_i - 0.96
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > -0.65\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -4.15\,\text{V}$. So this state holds for $\boxed{V_i > -0.65}$, where $V_o = -0.4V_i - 0.96$.

Step 5 — The other combinations

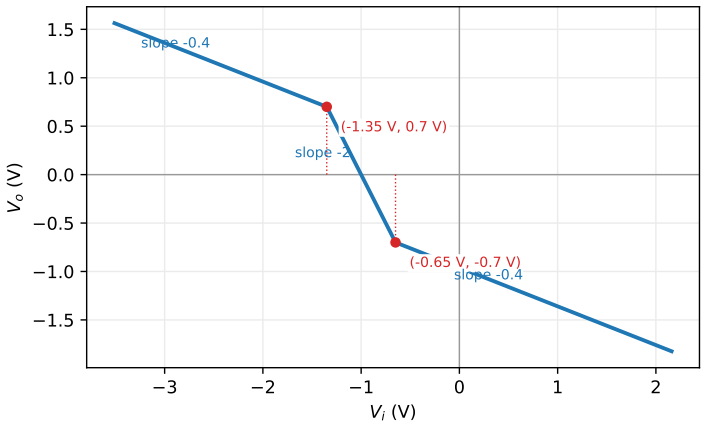
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 2.15$; $I_{D_2} > 0$ needs $V_i < -4.15$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

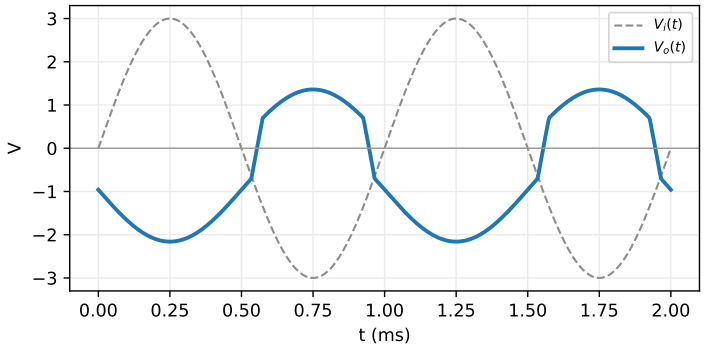
Put the valid states together:

$$V_o = \begin{cases} -0.4V_i + 0.16 & V_i < -1.35 \\ -2V_i - 2 & -1.35 < V_i < -0.65 \\ -0.4V_i - 0.96 & V_i > -0.65 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -1.35\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$; at $V_i = -0.65\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$. ✓



For $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -0.4V_i + 0.16 & V_i < -1.35 \\ -2V_i - 2 & -1.35 < V_i < -0.65 \\ -0.4V_i - 0.96 & V_i > -0.65 \end{cases}$$